

In this booklet, a number of problems are stated in the form of flow-charts.

1. A flow-chart is simply a "road map" by which you must find your way from one "town" to the next, and obey every instruction you come to.
2. Always follow the "road" downward, or to the right, unless an arrow indicates a different direction, or unless an instruction requires you to do otherwise.
3. When you come to a fork in the "road" there will be a question for you to answer; match your answer with the "signposts" on the branches leading out of the fork, and you will know which way to go.
4. When you are told to put a number into a box, it is understood that whatever number was previously in that box has just been erased.
5. All correct answers in this test are; whole, positive, numbers.
6. Mark up the flow-charts in any way you feel will help you. Use the margins, or the blank left-hand pages, for your "scratch" work. Do all your work in this booklet.
7. When you are finished, copy your answers into the spaces on this cover. Use extreme care in doing so, since this copy of your answers will be used in scoring the test.
8. Time will be limited to 1.5 hours

EXAMPLE A

Box #	1	2	3	4	5	6	7	8	9	10
	6	3	9	2	11	2	91	48	66	1

Start

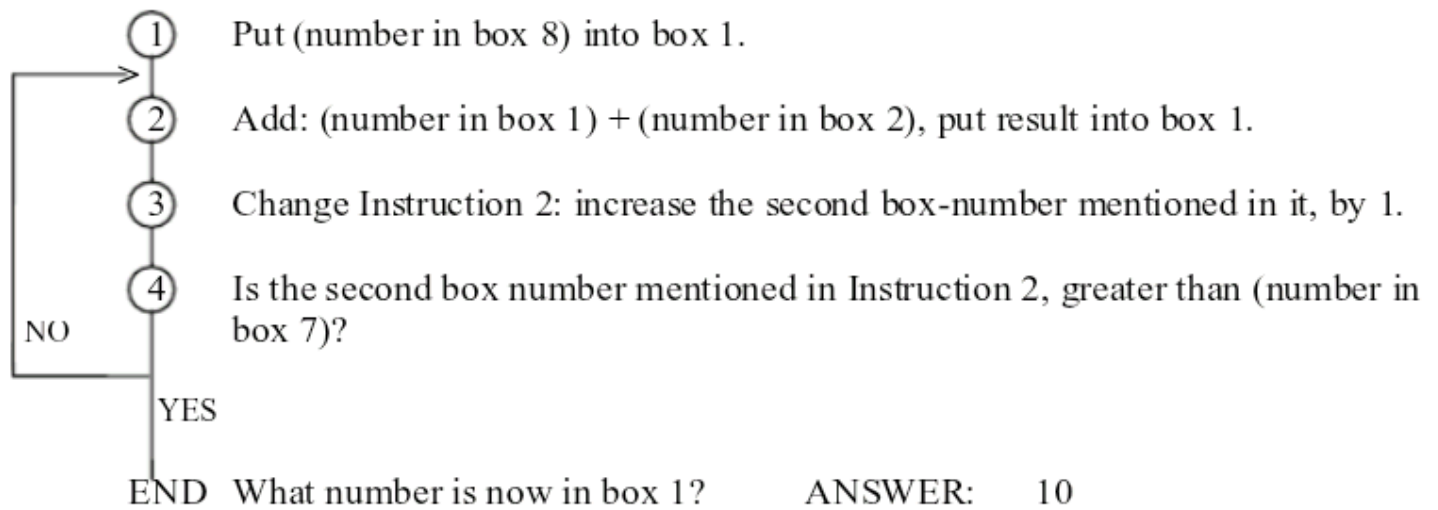
- ① Add: (number in box 4) + (number in box 2), put result into box 7.
- ② Add: (number in box 7) + (number in the box whose number is in box 6), put result into box 6.
- ③ Multiply: (number in box 6) X (number in box 1), put result into box 5

END What number is now in box 5? ANSWER: 48

EXAMPLE B:

Box #	1	2	3	4	5	6	7	8
	3	7	2	1	5	12	4	0

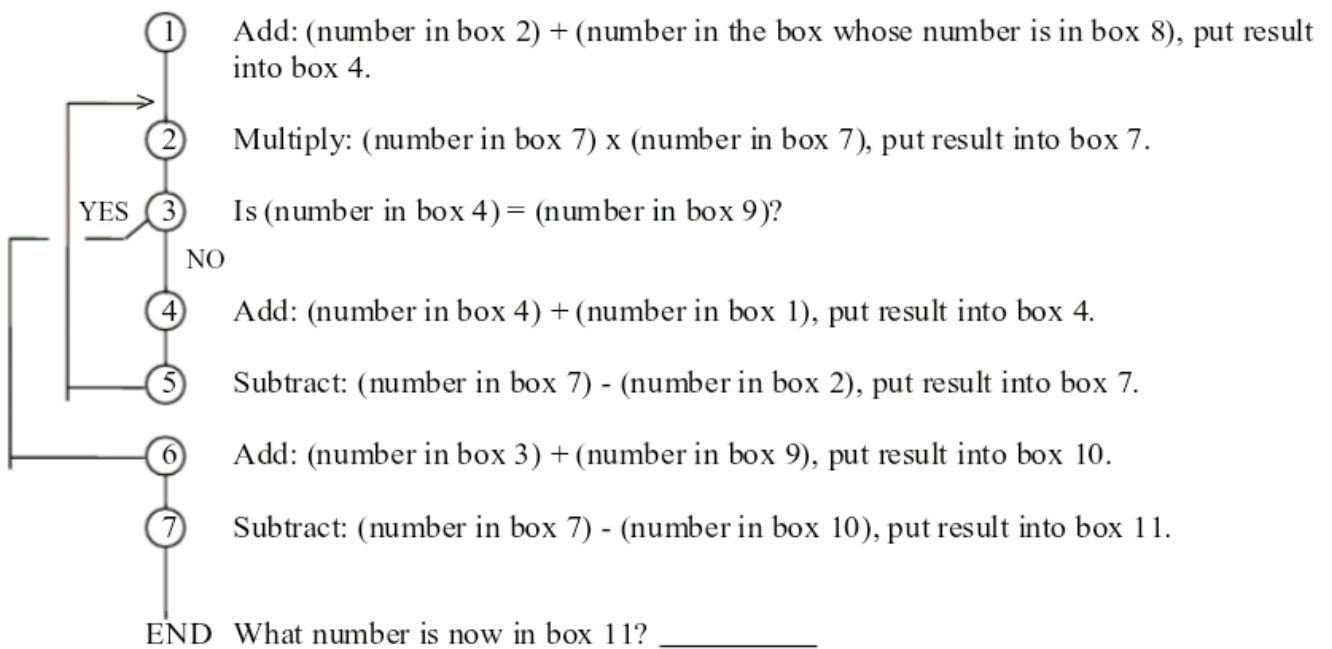
Start



QUESTION 1

Box #	1	2	3	4	5	6	7	8	9	10	11	12
	2	1	4	4	6	5	2	12	5	19	1	0

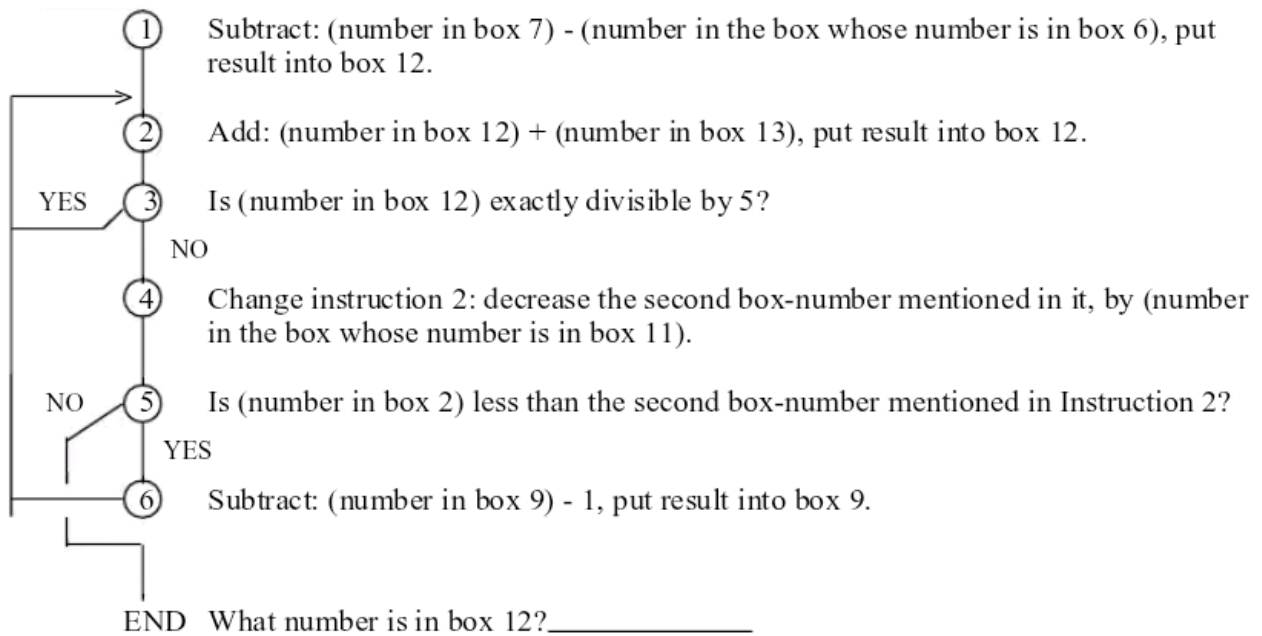
START



QUESTION 2

Box #	1	2	3	4	5	6	7	8	9	10	11	12	13
	9	8	5	2	11	3	5	12	7	-2	4	-6	6

START

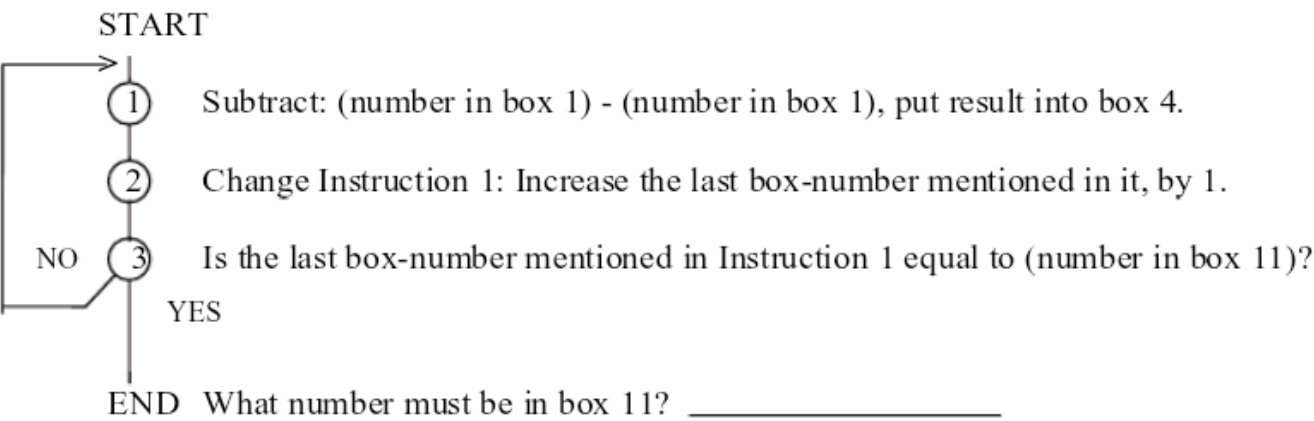


QUESTION 4

The purpose of the following flow-chart is to put a zero in each of the boxes: 4, 5 and 6.

In order to accomplish exactly this -- no more and no less -- what number must be in box 11?

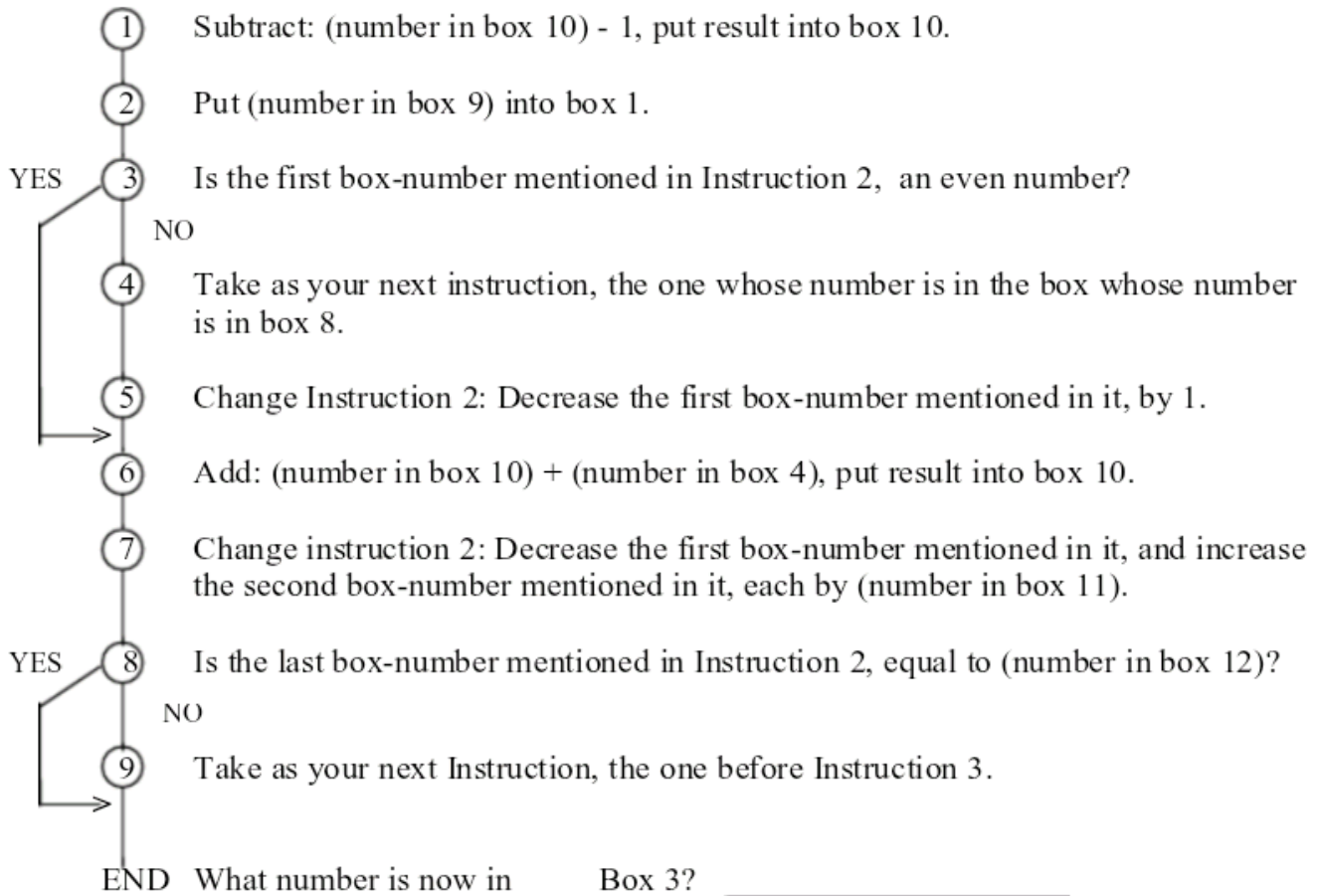
Box #	1	2	3	4	5	6	7	8	9	10	11	12
	7	9	2	2	8	4	1	-9	-3	6		4



QUESTION 3

Box #	1	2	3	4	5	6	7	8	9	10	11	12
	2	11	-9	3	-2	12	4	9	6	3	1	4

START



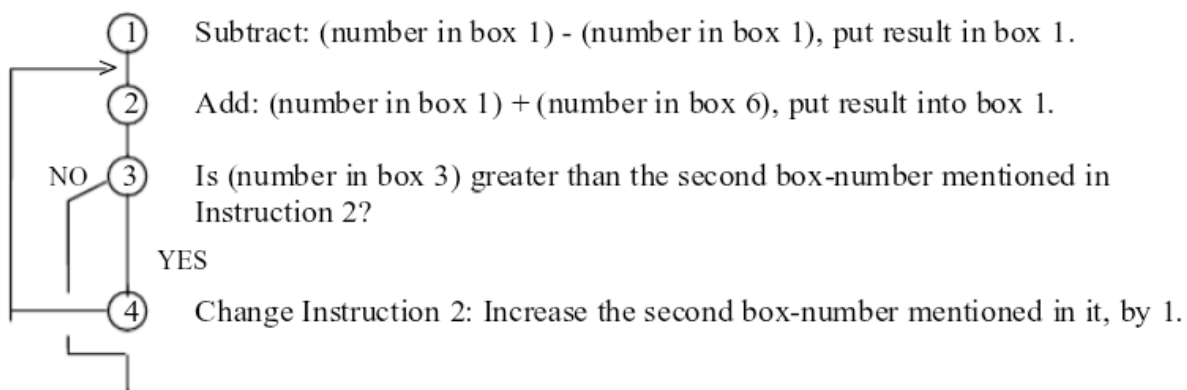
QUESTION 5

The purpose of the following flow-chart is to add up the numbers in boxes 6, 7, 8 and 9, and put the total in box 1.

In order to accomplish exactly this -- no more and no less -- what number must be in box 3?

Box #	1	2	3	4	5	6	7	8	9
	3	15		2	1	4	3	12	10

START

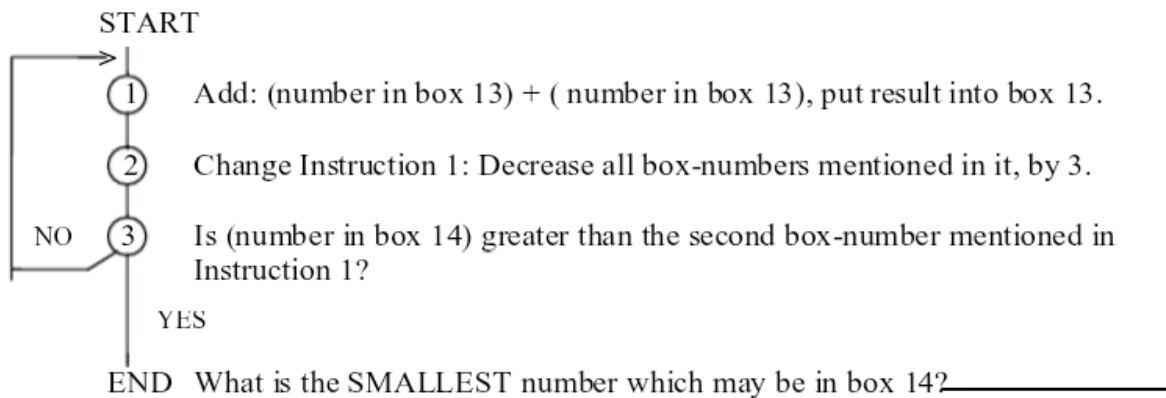


QUESTION 6

The purpose of the following flow-chart is to double the number in each of the boxes: 13, 10, 7 and 4.

In order to accomplish exactly this -- no more and no less -- what is the smallest number which may be in box 14?

Box #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

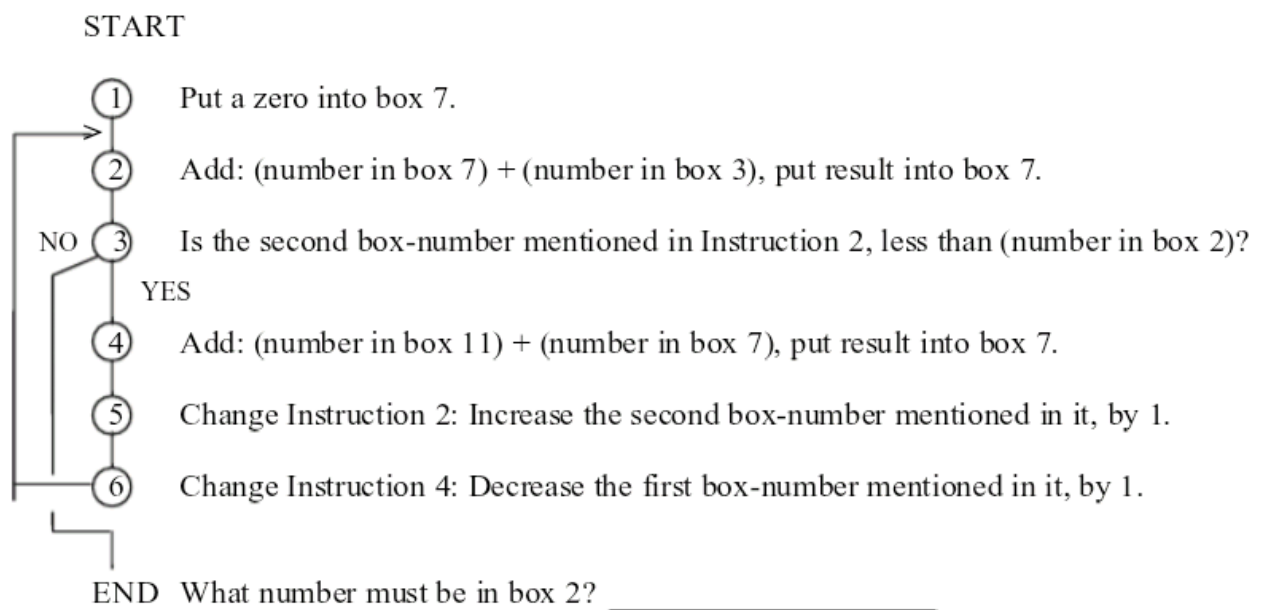


QUESTION 7

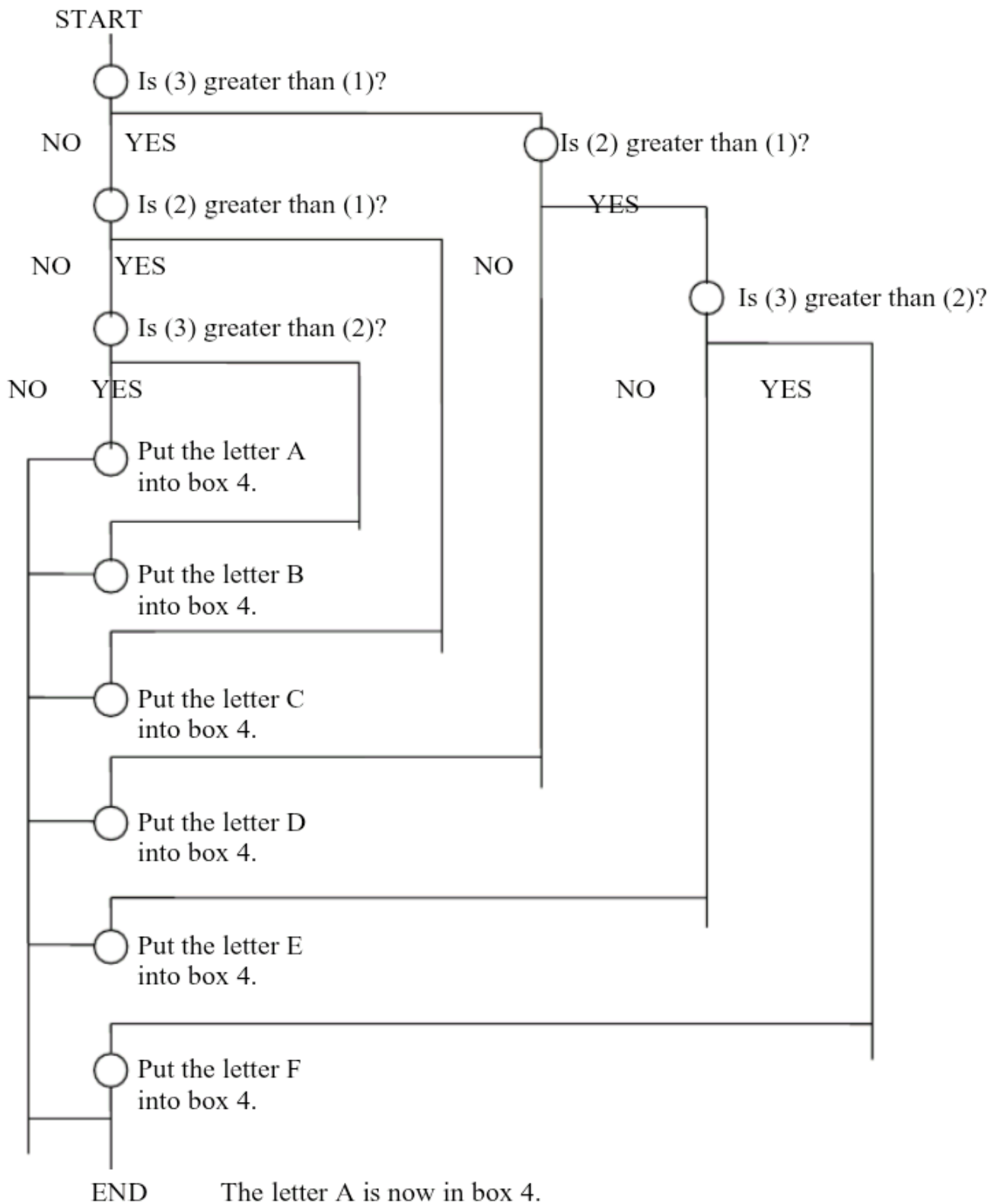
The purpose of the following flow-chart is to add up the numbers in boxes 3, 4, 5, 6, 9, 10 and 11, and to put the total in box 7.

In order to accomplish exactly this -- no more and no less -- what number must be in box 2?

Box #	1	2	3	4	5	6	7	8	9	10	11	12	13



QUESTION 8

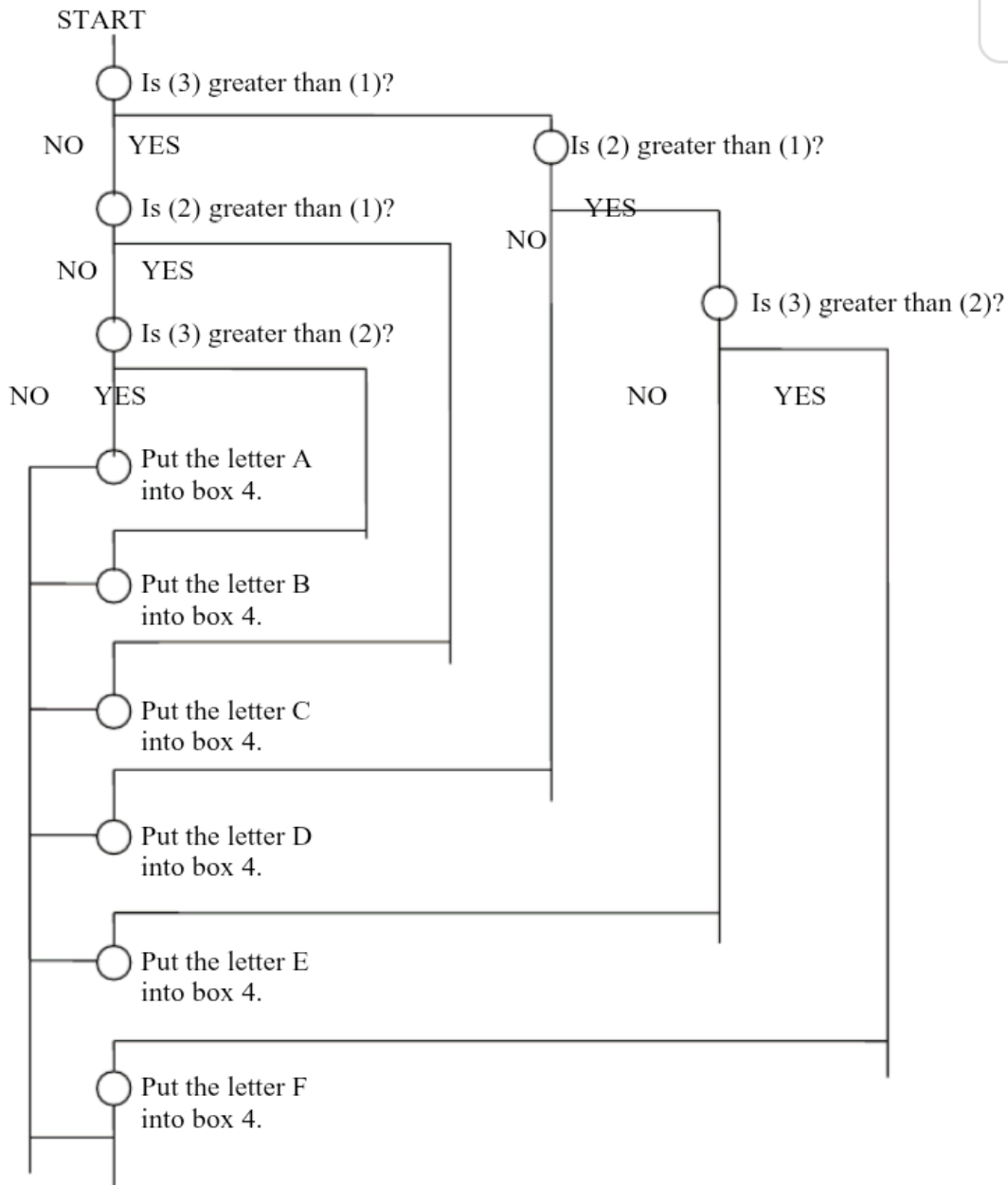


Which of the boxes 1, 2, 3 contains... the largest number? _____

the smallest number? _____

QUESTION 9

14 / 17

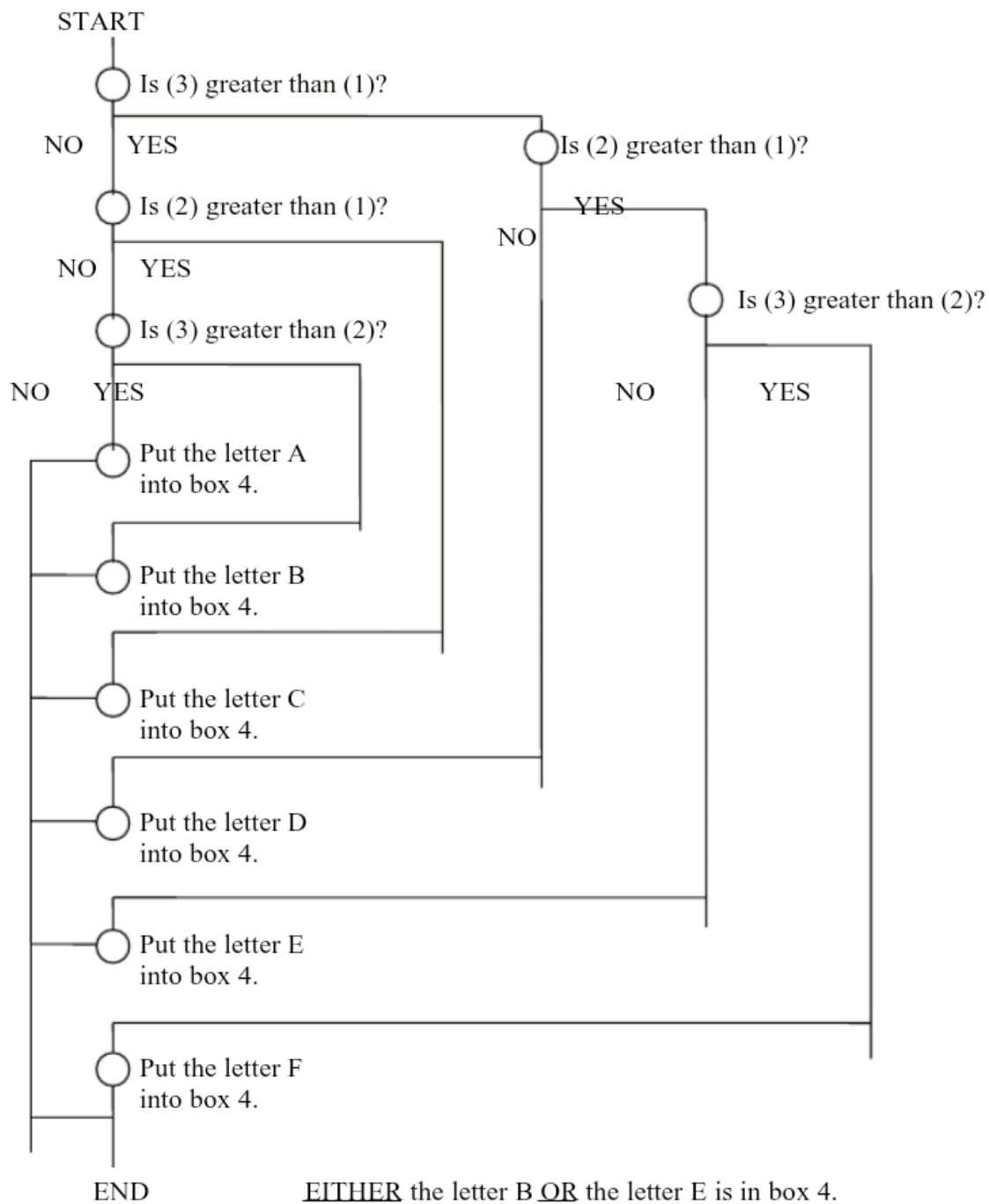


END EITHER the letter C OR the letter F is in box 4.

Which of the boxes 1, 2, 3 CANNOT POSSIBLY contain the largest number? _____

the smallest number? _____

QUESTION 10



EITHER the letter B OR the letter E is in box 4.

Which of the boxes 1, 2, 3 CANNOT POSSIBLY contain the largest number? _____
 the smallest number? _____

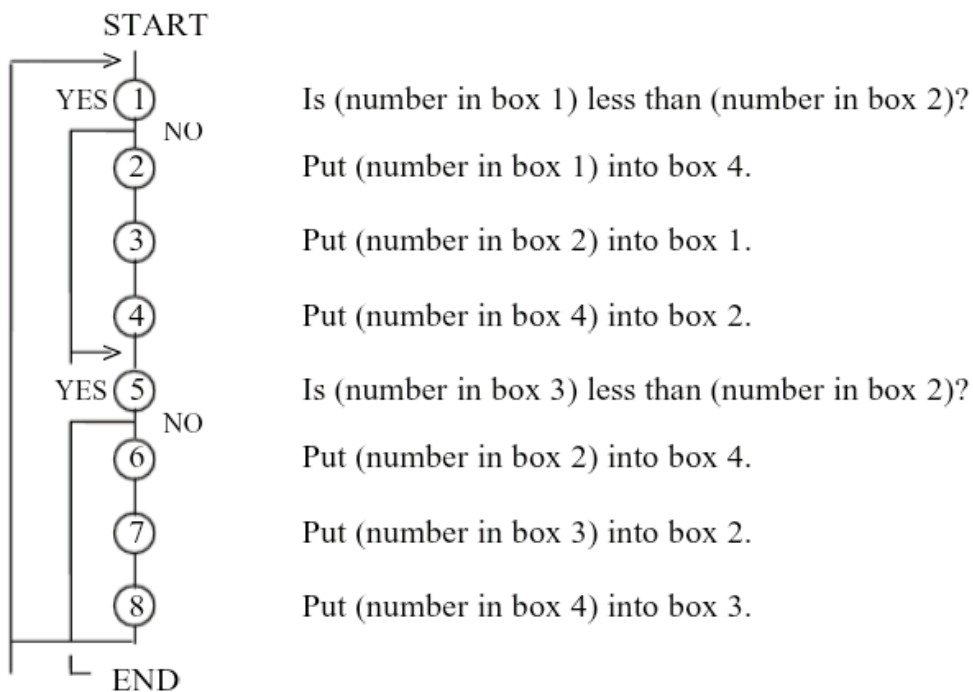
QUESTION 11

The object of the following flow-chart is to rearrange the numbers in boxes 1, 2 and 3 in such a fashion that, at the end:

Box 3 will always contain the largest number,
Box 1 will always contain the smallest number,

However, this flow-chart is INCORRECT. As it stands, it will not accomplish the desired purpose.

Which one instruction must be changed to correct the flow-chart?



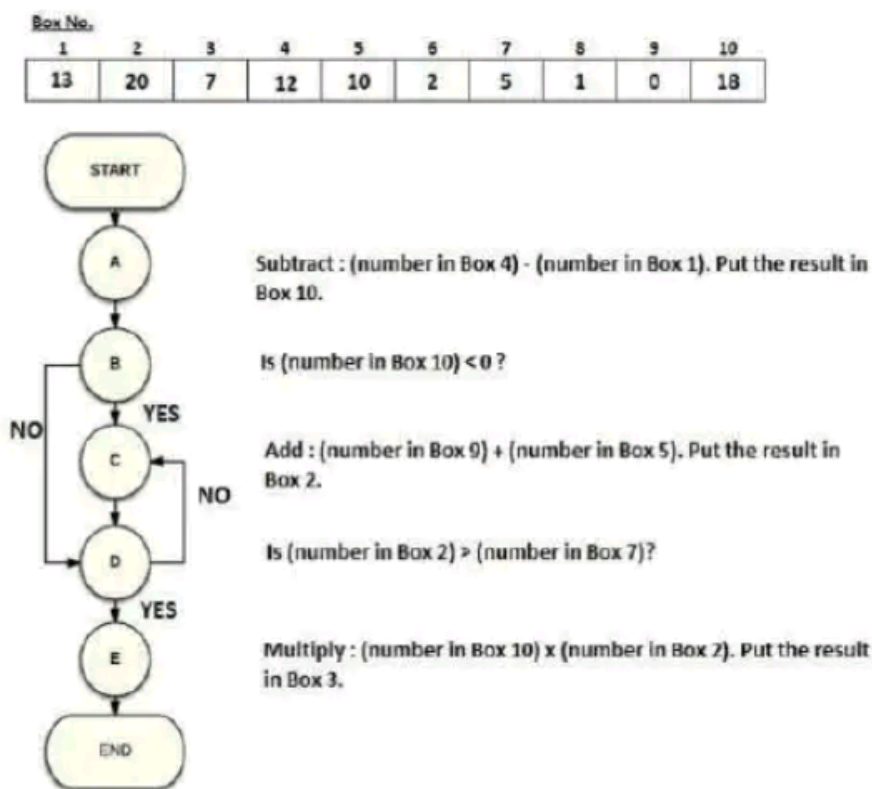
In order to correct this flow-chart, we must change Instruction No. _____

So that the first box-number mentioned in it is box-number..... _____

And the second box-number mentioned in it is box-number..... _____

1. Study the flow chart give below and the questions that follow

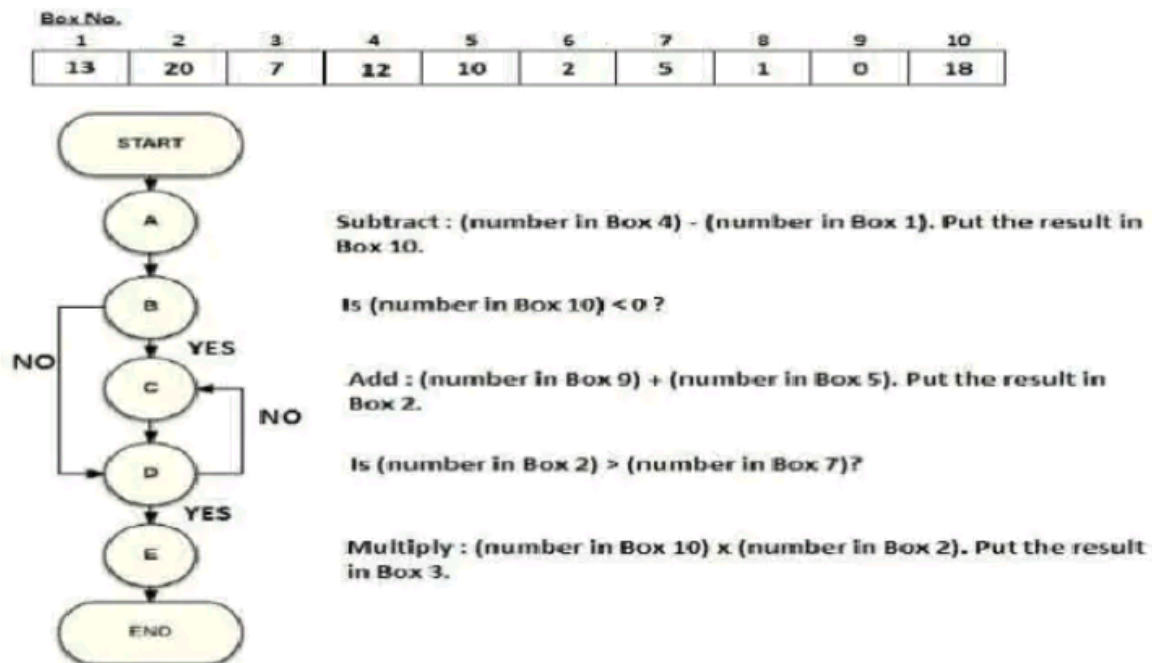
Question: At the end of the flow chart the number placed in which of the following boxes will remain unchanged



A. Box 3 **B.** Box 10 **C.** Box 2 **D.** Box 7

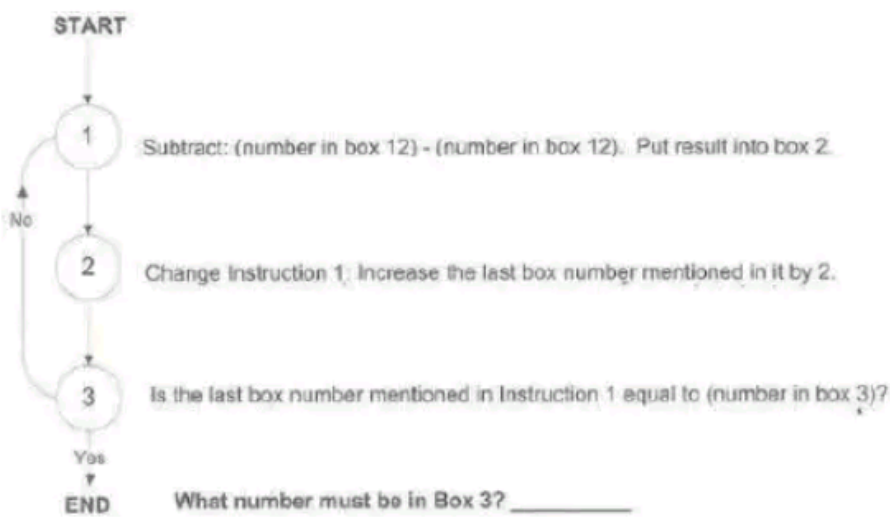
2. Is the following statement True or False Statement:

If the condition in Step C updates the value in Box 3 instead of Box 2, then the flow chart will enter into infinite loop



3. Study the flow chart give below and the questions that follow.
The purpose of the following chart is to put a zero in each of the boxes 2,4 and 6. In order to accomplish exactly this - no more and no less - Which number must be in box 3?

Box No.	1	2	3	4	5	6	7	8	9	10	11	12
	7	6		1	6	10	4	3	2	14	8	9



- A. 0 B. 6 C. 10 D. 3 E. 8